

Name: _____

Date: _____

ALGEBRAIC FRACTIONS, (SIMPLIFYING ALGEBRAIC FRACTIONS)

LO: I can simplify algebraic fractions by factorising and cancelling common factors.

Simplifying algebraic fractions

To **simplify** an algebraic fraction, factorise the numerator and denominator, then **cancel common factors**.

Strategy: Look for common numerical factors first, then factorise any quadratic or linear expressions.

Quick examples

●	$4x/8 = x/2$	divide by 4
●	$x^2+xy/x = x + 1$	factorise and cancel x
●	$x^2-4/x+2 = x - 2$	difference of two squares
●	$x+3/x+5 = 3/5$	cannot cancel x from a sum
●	$x^2+1/x = x + 1$	x^2+1 does not factorise over x

Key idea

Factorise numerator and denominator completely, then cancel any factors that appear in both.

$$x^2 - 4/x + 2 = \frac{(x+2)(x-2)}{x+2} = x - 2$$

Cancel the common factor (x + 2)

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – common numerical factor

Simplify $6x/9$

HCF of 6 and 9 is 3

Divide both by 3

$$6x/9 = 2x/3$$

$$2x/3$$

Divide numerator and denominator by their HCF.

Example 2 – common variable factor

Simplify $x^2 + 3x/x$

Factorise numerator: $x(x + 3)$

Rewrite: $x(x + 3)/x$

Cancel x: $= x + 3$

$$= x + 3$$

Factorise the numerator to reveal the common factor.

Example 3 – difference of two squares

Simplify $x^2 - 9/x + 3$

Factorise: $x^2 - 9 = (x + 3)(x - 3)$

Rewrite: $(x+3)(x-3)/x+3$

Cancel (x + 3): $= x - 3$

$$= x - 3$$

Recognise the difference of two squares pattern: $a^2 - b^2 = (a+b)(a-b)$.

Example 4 – quadratic numerator

Simplify $x^2 + 5x + 6/x + 2$

Factorise: $x^2 + 5x + 6 = (x + 2)(x + 3)$

Rewrite: $(x+2)(x+3)/x+2$

Cancel (x + 2): $= x + 3$

$$= x + 3$$

Factorise the quadratic to find the common factor with the denominator.

YOUR TURN – PRACTICE (deliberate practice)

1. Common numerical factors

Simplify each fraction.

- a) $\frac{8x}{12}$
- b) $\frac{15x}{5}$
- c) $\frac{10x^2}{2x}$
- d) $\frac{12x^2}{4x}$

2. Factorising the numerator

Factorise and simplify.

- a) $\frac{x^2 + 2x}{x}$
- b) $\frac{3x + 6}{3}$
- c) $\frac{x^2 - 5x}{x}$
- d) $\frac{2x^2 + 4x}{2x}$

3. Difference of two squares

Simplify using DOTS.

- a) $\frac{x^2 - 4}{x + 2}$
- b) $\frac{x^2 - 16}{x - 4}$
- c) $\frac{x^2 - 25}{x + 5}$
- d) $\frac{x^2 - 1}{x + 1}$

4. Quadratic numerator

Factorise and simplify.

- a) $\frac{x^2 + 4x + 3}{x + 1}$
- b) $\frac{x^2 + 7x + 12}{x + 3}$
- c) $\frac{x^2 - x - 6}{x + 2}$
- d) $\frac{x^2 + 2x - 8}{x - 2}$

5. Mixed simplification

Simplify each algebraic fraction fully.

- a) $\frac{5x + 10}{x + 2}$
- b) $\frac{x^2 - 9}{x^2 + 6x + 9}$
- c) $\frac{2x^2 - 8}{2x + 4}$
- d) $\frac{x^2 + x - 12}{x^2 - 16}$
- e) $\frac{x^2 - 6x + 9}{x^2 - 9}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $\frac{x^2 - 4}{x + 2} = x - 2$ ☐

Correct answer: _____

Reason: _____

b) $\frac{x + 3}{x + 5} = \frac{3}{5}$ ☐

Correct answer: _____

Reason: _____

c) $\frac{x^2 + 3x}{x} = x + 3$ ☐

Correct answer: _____

Reason: _____

d) $\frac{x^2 + 4}{x + 2} = x + 2$ ☐

Correct answer: _____

Reason: _____

e) You must factorise before cancelling ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Simplify $\frac{x^2 + 2x - 15}{x^2 - 9}$ fully. State any values of x for which the fraction is undefined.

Expression: _____

Simplified: _____

b) Show that $\frac{2x^2 + 7x + 3}{2x^2 + 5x + 2}$ simplifies to $\frac{x + 3}{x + 2}$. Explain each factorisation step.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

